EXPERIMENT NO: 03

1. ROLL-UP(Total sales by cuisine)

```
SELECT
    R.Cuisine,
    SUM(F.Order_Value) AS Total_Sales
FROM FACT_DELIVERY F
JOIN DIM_RESTAURANT R
ON F.Restaurant_Key = R.Restaurant_Key
GROUP BY R.Cuisine;
```

```
+-----------+--------------+
| Cuisine   | Total_Sales |
+-----------+--------------+
| Italian   |          650 |
| Indian    |          420 |
| Japanese |          980 |
+-----------+--------------+
```

2. DRILL-DOWN (Cuisine → Restaurant)

```
SELECT
    R.Cuisine,
    R.Restaurant_Name,
    SUM(F.Order_Value) AS Total_Sales
FROM FACT_DELIVERY F
JOIN DIM_RESTAURANT R
ON F.Restaurant_Key = R.Restaurant_Key
GROUP BY
    R.Cuisine,
    R.Restaurant_Name
ORDER BY
    R.Cuisine,
    R.Restaurant_Name;
```

```
+----------+-----------------+-------------+
| Cuisine  | Restaurant_Name | Total_Sales |
+----------+-----------------+-------------+
| Indian   | Spice Villa     |         420 |
| Italian  | Pizza Hub       |         650 |
| Japanese | Sushi House     |         980 |
+----------+-----------------+-------------+
```

3. SLICE (Only Indian Restaurants)

```
SELECT
   R.Restaurant_Name,
   R.Cuisine,
   F.Order_Value
FROM FACT_DELIVERY F
JOIN DIM_RESTAURANT R
ON F.Restaurant_Key = R.Restaurant_Key
WHERE R.Cuisine = 'Indian';
```

```
+-----------------+---------+-------------+
| Restaurant_Name | Cuisine | Order_Value |
+-----------------+---------+-------------+
| Spice Villa     | Indian  |         420 |
+-----------------+---------+-------------+
```

4. DICE (Restaurants in Mumbai with Rating ≥ 4.5)

```
SELECT
   R.Restaurant_Name,
   R.City,
   R.Rating,
   F.Order_Value
FROM FACT_DELIVERY F
JOIN DIM_RESTAURANT R
ON F.Restaurant_Key = R.Restaurant_Key
WHERE R.City = 'Mumbai'
  AND R.Rating >= 4.5;
```

```
+-----------------+--------+--------+-------------+
| Restaurant_Name | City   | Rating | Order_Value |
+-----------------+--------+--------+-------------+
| Pizza Hub       | Mumbai |    4.5 |         650 |
+-----------------+--------+--------+-------------+
```

5. PIVOT (Total Order Value)

```
SELECT *
FROM (
   SELECT R.City, F.Order_Value
   FROM FACT_DELIVERY F
   JOIN DIM_RESTAURANT R
     ON F.Restaurant_Key = R.Restaurant_Key
)
PIVOT (
   SUM(Order_Value)
   FOR City IN ('Mumbai' AS MUMBAI,
                'Delhi' AS DELHI,
                'Bangalore' AS BANGALORE)
);
```

```
+---------+-------+-----------+
| MUMBAI  | DELHI | BANGALORE |
+---------+-------+-----------+
| 650     | 420   | 980       |
+---------+-------+-----------+
```